ADVANCED QUANTUM MECHANICS

Fundamentals of quantum mechanics

TECHNISCHE UNIVERSITÄT DARMSTADT

Claudiu Genes, Institute for Applied Physics; TU Darmstadt, winter semester 2025/2026

LECTURE 5: Spin-orbit coupling. Time dependent perturbation theory. Fermi's golden rule.

We will begin by analyzing the fine-structure splitting of the hydrogen atom, which arises from the fact that the electron possesses an additional intrinsic property — its spin. We will proceed as follows:

- We introduce the spin–orbit interaction Hamiltonian and explain the necessity of using a basis defined by the *total angular momentum* $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$.
- Apply time-independent perturbation theory to the spin–orbit coupling in order to evaluate the corresponding energy shifts.
- Introduce time-dependent perturbation theory to describe resonant transitions between atomic orbitals induced by external fields.
- Derive *Fermi's golden rule*, one of the most powerful and widely used results in quantum mechanics.

The spin orbit Hamiltonian. Application of degenerate time independent perturbation theory.

The spin–orbit Hamiltonian comes from the from relativistic Dirac equation which predicts that the electron is described by a spinor, a two component state vector. This means that the electron has a spin magnetic moment which interacts with the magnetic field generated by its own orbital motion around the nucleus. This interaction causes a splitting of atomic energy levels that would otherwise be degenerate, and it is a fundamental concept in atomic physics and solid-state physics. The coupling is stronger for atoms with heavy nuclei, as the effect is proportional to the nuclear charge.

Let us describe here the procedure to treat the spin orbit coupling, before we proceed to analyze each aspect separately. We start with the Hamiltonian

$$H_{\text{so}} = A\,\frac{1}{r^3}\,\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}, \qquad \text{with} \qquad A \equiv \frac{Ze^2}{8\pi\varepsilon_0\, m_e^2 c^2}.$$

For hydrogen $Z = 1$. In order to find how this perturbation changes the states of the system we proceed as follows:

(i) **Add spin in our description** Diagonalize $\hat{H}_0 = \dfrac{\hat{\mathbf{p}}^2}{2m_e} + V(r)$ with the hydrogenic basis $|n, \ell, m_\ell, m_s\rangle$ where $m_s = \pm 1/2$.

(ii) **Angular-momentum addition.** The easiest approach is to define coupled states $|n, \ell, s;\, j, m_j\rangle$ with $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$ and

$$\hat{\mathbf{L}}\cdot\hat{\mathbf{S}} = \tfrac{1}{2}\big(\hat{\mathbf{J}}^2 - \hat{\mathbf{L}}^2 - \hat{\mathbf{S}}^2\big).$$

(iii) **Perturbative matrix element.** In first-order perturbation theory,

$$\langle\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}\rangle = \frac{\hbar^2}{2}\Big[j(j{+}1) - \ell(\ell{+}1) - s(s{+}1)\Big], \qquad s = \tfrac{1}{2}.$$

(iv) **Radial average.** For hydrogenic states with $\ell \geq 1$,

$$\langle r^{-3} \rangle_{n\ell} = \frac{Z^3}{a_0^3} \frac{1}{n^3 \, \ell(\ell + \frac{1}{2})(\ell + 1)}.$$

For $\ell = 0$, $\hat{\mathbf{L}} = 0 \Rightarrow \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = 0$; the $1s$ fine structure arises from the Darwin and relativistic kinetic terms.

For all other states we have:

$$\boxed{\Delta E_{\text{so}}(n, \ell, j) = A \, \langle r^{-3} \rangle_{n\ell} \, \frac{\hbar^2}{2} \Big[j(j{+}1) - \ell(\ell{+}1) - \tfrac{3}{4} \Big], \qquad (\ell \geq 1)}$$

We will prove this in the following.

The spin of the electron

For an electron, the spin quantum number is $s = \frac{1}{2}$, with two possible projections $m_s = \pm\frac{1}{2}$, corresponding to the states

$$|{\uparrow}\rangle = |s = \tfrac{1}{2}, m_s = +\tfrac{1}{2}\rangle, \qquad |{\downarrow}\rangle = |s = \tfrac{1}{2}, m_s = -\tfrac{1}{2}\rangle$$

The spin operators satisfy the same angular momentum algebra, $[S_i, S_j] = i\hbar \, \epsilon_{ijk} S_k$, and the eigenvalues of $\hat{\mathbf{S}}^2$ are $s(s+1)\hbar^2$ and of $\hat{S}_z$ are $\pm\hbar/2$. One can use the Pauli matrices:

$$S_x = \frac{\hbar}{2}\sigma_x, \qquad S_y = \frac{\hbar}{2}\sigma_y, \qquad S_z = \frac{\hbar}{2}\sigma_z.$$

Since the unperturbed Hamiltonian of the hydrogen atom,

$$\hat{H}_0 = \frac{\hat{\mathbf{p}}^2}{2m_e} + V(\hat{r}),$$

acts only on spatial degrees of freedom, it commutes with all spin operators,

$$[\hat{H}_0, \hat{\mathbf{S}}] = 0.$$

Therefore, spin is conserved under $\hat{H}_0$. Each eigenstate of the spatial part can be written as $|n, \ell, m_\ell\rangle$, and the full state including spin becomes

$$|n, \ell, m_\ell, m_s\rangle = |n, \ell, m_\ell\rangle \otimes |m_s\rangle.$$

This doubles the degeneracy of each orbital, yielding a total of $2(2\ell + 1)$ states for given (n, ℓ), and explains why each orbital can accommodate two electrons with opposite spin orientations. For each orbital characterized by (n, ℓ, m_ℓ), the presence of spin allows two possible spin orientations, $m_s = \pm\frac{1}{2}$.

In the position representation, the electron is described by a *spinor wavefunction* (a column vector with two components)

$$\psi_{n\ell m_\ell m_s}(\mathbf{r}) = \langle \mathbf{r} | n, \ell, m_\ell, m_s \rangle . = R_{n\ell}(r) \, Y_{\ell m_\ell}(\theta, \phi) \, \chi_{m_s}$$

where χ_{m_s} is the spinor

$$\chi_{+\frac{1}{2}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \chi_{-\frac{1}{2}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

According to **Pauli's exclusion principle**, no two electrons in an atom can occupy the same set of quantum numbers (n, ℓ, m_ℓ, m_s). Hence:

- The $1s$ orbital $(n = 1, \ell = 0)$ can hold **two electrons**: one spin-up and one spin-down.
- The $2s$ orbital $(n = 2, \ell = 0)$ can also hold **two electrons**.
- The $2p$ subshell $(n = 2, \ell = 1)$ has three spatial orbitals $(m_\ell = -1, 0, +1)$, each accommodating two spin orientations, giving a total of **six electrons**.

- The $3s$ orbital ($n=3, \ell=0$) holds **two electrons**.
- The $3p$ subshell ($n=3, \ell=1$) holds **six electrons**.
- The $3d$ subshell ($n=3, \ell=2$) has five spatial orbitals ($m_\ell = -2, -1, 0, +1, +2$), each with two spin states, for a total of **ten electrons**.

Altogether, this explains the shell structure

$$1s^2, \quad 2s^2\,2p^6, \quad 3s^2\,3p^6\,3d^{10}, \quad \text{and so on.}$$

Addition of angular momenta

All energy levels with a given n are degenerate in energy. Which means that we have to apply degenerate perturbation theory, which means in the first step to diagonalize the Hamiltonian in such a way to get rid of the degeneracy. This is automatically taken care of by working in the collective angular momentum basis.

On the combined Hilbert space

$$\mathcal{H} = \mathcal{H}_{\text{orbital}} \otimes \mathcal{H}_{\text{spin}},$$

the total angular momentum operator is written as

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} \otimes \mathbb{1}_{\text{spin}} + \mathbb{1}_{\text{orbital}} \otimes \hat{\mathbf{S}},$$

where $\mathbb{1}_{\text{orbital}}$ acts on the orbital part (of dimension $2\ell+1$ for a given ℓ) and $\mathbb{1}_{\text{spin}} = \mathbb{1}_2$ acts on the spin-$\frac{1}{2}$ space. This becomes the relevant conserved quantity.

Once the spin–orbit term $\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$ is included in the Hamiltonian, $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$ no longer commute with $\hat{H}$ individually, but $\hat{\mathbf{J}}$ does:

$$[\hat{H}, \hat{\mathbf{J}}] = 0, \qquad [\hat{H}, \hat{\mathbf{L}}] \neq 0, \qquad [\hat{H}, \hat{\mathbf{S}}] \neq 0.$$

Hence, the correct quantum numbers are those of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$:

$$\hat{\mathbf{J}}^2 \left| n, \ell, s, j, m_j \right\rangle = \hbar^2 j(j+1) \left| n, \ell, s, j, m_j \right\rangle, \qquad \hat{J}_z \left| n, \ell, s, j, m_j \right\rangle = \hbar m_j \left| n, \ell, s, j, m_j \right\rangle.$$

For given ℓ and s, the possible total angular momenta are

$$j = \ell + s,\ \ell + s - 1,\ \ldots,\ |\ell - s|.$$

For an electron with $s = \frac{1}{2}$ this gives two possible values,

$$j = \ell + \tfrac{1}{2}, \qquad j = \ell - \tfrac{1}{2}.$$

The eigenstates of total angular momentum can be expanded in the product basis using Clebsch–Gordan coefficients:

$$\left| \ell, s; j, m_j \right\rangle = \sum_{m_\ell, m_s} C^{j\,m_j}_{\ell\,m_\ell,\,s\,m_s} \left| \ell, m_\ell \right\rangle \left| s, m_s \right\rangle,$$

where the coefficients $C^{j\,m_j}_{\ell\,m_\ell,\,s\,m_s}$ satisfy the rule $m_j = m_\ell + m_s$.

Example: for $\ell = 0$ (an s-orbital) and $s = \frac{1}{2}$, we have only $j = \frac{1}{2}$ and $m_j = \pm\frac{1}{2}$. The total states are

$$\left| \ell = 0, s = \tfrac{1}{2}; j = \tfrac{1}{2}, m_j = +\tfrac{1}{2} \right\rangle = C^{\frac{1}{2}\,\frac{1}{2}}_{0\,0,\,\frac{1}{2}\,\frac{1}{2}} \left| \ell = 0, m_\ell = 0 \right\rangle \left| \uparrow \right\rangle = \left| \ell = 0, m_\ell = 0 \right\rangle \left| \uparrow \right\rangle,$$

$$\left| \ell = 0, s = \tfrac{1}{2}; j = \tfrac{1}{2}, m_j = -\tfrac{1}{2} \right\rangle = C^{\frac{1}{2}\,-\frac{1}{2}}_{0\,0,\,\frac{1}{2}\,-\frac{1}{2}} \left| \ell = 0, m_\ell = 0 \right\rangle \left| \downarrow \right\rangle = \left| \ell = 0, m_\ell = 0 \right\rangle \left| \downarrow \right\rangle.$$

Since $\ell = 0$ implies $m_\ell = 0$, each state corresponds directly to one spin orientation; the Clebsch–Gordan coefficients are 1 in this case.

Let us now notice a couple of things which will be useful:

(i) We can easily compute the angular momentum part:
$$\hat{\mathbf{J}}^2 = (\hat{\mathbf{L}} + \hat{\mathbf{S}})^2 = \hat{\mathbf{L}}^2 + \hat{\mathbf{S}}^2 + 2\,\hat{\mathbf{L}}\cdot\hat{\mathbf{S}} \implies \hat{\mathbf{L}}\cdot\hat{\mathbf{S}} = \tfrac{1}{2}\big(\hat{\mathbf{J}}^2 - \hat{\mathbf{L}}^2 - \hat{\mathbf{S}}^2\big).$$
Since $[L_i, S_j] = 0$, the cross terms commute and combine symmetrically.

(ii) We can also find the basis for diagonalization easily, as the spin–orbit perturbation $\Delta H \propto \hat{\mathbf{L}}\cdot\hat{\mathbf{S}}$ is diagonal in the simultaneous eigenbasis of $\hat{\mathbf{J}}^2$, $\hat{\mathbf{L}}^2$, $\hat{\mathbf{S}}^2$, and $\hat{J}_z$:
$$|n, \ell, s, j, m_j\rangle\,.$$
The corresponding eigenvalues are
$$\hat{\mathbf{L}}^2 \to \hbar^2\ell(\ell+1), \qquad \hat{\mathbf{S}}^2 \to \hbar^2 s(s+1), \qquad \hat{\mathbf{J}}^2 \to \hbar^2 j(j+1), \qquad \hat{J}_z \to \hbar m_j.$$

(iii) Now we can compute the expectation value
$$\langle j, \ell, s, m_j|\ \hat{\mathbf{L}}\cdot\hat{\mathbf{S}}\ |j, \ell, s, m_j\rangle = \frac{\hbar^2}{2}\big[j(j+1) - \ell(\ell+1) - s(s+1)\big].$$
For an electron with $s = \frac{1}{2}$, there are two possible total angular momenta:
$$j = \ell + \tfrac{1}{2} \qquad \text{and} \qquad j = \ell - \tfrac{1}{2},$$
corresponding to the fine-structure splitting of energy levels.

Spin–orbit coupling as a perturbation

Now let's put it all together. In the basis of the total angular momentum the Hamiltonian is diagonal and the first-order energy shift is $|n, \ell, s; j, m_j\rangle$ is
$$\Delta E_{\rm so}(n,\ell,j) = \langle n, \ell, s; j, m_j|\, \hat{H}_{\rm so}\, |n, \ell, s; j, m_j\rangle = A\ \underbrace{\langle n, \ell|\ r^{-3}\ |n, \ell\rangle}_{\text{radial}}\ \underbrace{\langle \ell, s; j, m_j|\, \hat{\mathbf{L}}\cdot\hat{\mathbf{S}}\, |\ell, s; j, m_j\rangle}_{\text{angular}}.$$
The angular factor we already computed above. For the radial factor, we use $|n, \ell, m_\ell\rangle = |R_{n\ell}\rangle \otimes |Y_{\ell m_\ell}\rangle$ with
$$\langle \mathbf{r}|\ n, \ell, m_\ell\rangle = R_{n\ell}(r)\, Y_{\ell m_\ell}(\Omega), \qquad \int_0^\infty dr\, r^2 |R_{n\ell}(r)|^2 = 1, \qquad \int d\Omega\, |Y_{\ell m_\ell}|^2 = 1.$$
Then
$$\langle n, \ell|\ r^{-3}\ |n, \ell\rangle = \int d^3\mathbf{r}\, |R_{n\ell}(r)|^2 |Y_{\ell m_\ell}(\Omega)|^2\, r^{-3} = \int_0^\infty dr\, r^2 |R_{n\ell}(r)|^2\, r^{-3} = \int_0^\infty dr\, \frac{|R_{n\ell}(r)|^2}{r}.$$
Using the standard radial function
$$R_{n\ell}(r) = \frac{2}{n^2}\sqrt{\frac{(n-\ell-1)!}{(n+\ell)!}}\, \left(\frac{2Zr}{na_0}\right)^\ell e^{-Zr/(na_0)}\, L_{n-\ell-1}^{(2\ell+1)}\!\left(\frac{2Zr}{na_0}\right),$$
set $\rho = \frac{2Zr}{na_0}$ so that $r = \frac{na_0}{2Z}\rho$ and $dr = \frac{na_0}{2Z}d\rho$. Then
$$\langle n, \ell|\ r^{-3}\ |n, \ell\rangle = \left(\frac{2Z}{na_0}\right)\int_0^\infty d\rho\, \frac{|R_{n\ell}(r(\rho))|^2}{\rho}.$$
Carrying out the Laguerre-weighted integral (orthogonality with weight $\rho^{2\ell+1}e^{-\rho}$) yields, for $\ell \geq 1$,
$$\boxed{\langle r^{-3}\rangle_{n\ell} = \frac{Z^3}{a_0^3}\, \frac{1}{n^3\, \ell\big(\ell + \frac{1}{2}\big)(\ell+1)}}.$$
For $\ell = 0$, the integral of r^{-3} is formally divergent at the origin, but $\hat{\mathbf{L}} = 0$ implies $\hat{\mathbf{L}}\cdot\hat{\mathbf{S}} = 0$, so the spin–orbit shift vanishes in $1s$; the $1s$ fine structure comes instead from the Darwin and relativistic kinetic terms.

The final expression is what we have listed in the beginning of the lecture.

Degenerate vs. non-degenerate perturbation theory. Let us stress again this point. The unperturbed Coulomb Hamiltonian $\hat{H}_0$ has n-fold degeneracy across different ℓ (same n). Within a fixed (n, ℓ) manifold, there is further $2(2\ell+1)$ degeneracy from m_ℓ and m_s. The operator $\hat{H}_{\rm so} \propto \hat{\mathbf{L}}\cdot\hat{\mathbf{S}}$ commutes with $\hat{\mathbf{J}}^2$ and $\hat{J}_z$, so in the *uncoupled* basis $|n, \ell, m_\ell, m_s\rangle$ one should use *degenerate* perturbation theory and diagonalize in the (m_ℓ, m_s) subspace. Choosing the *coupled* basis $|n, \ell, s; j, m_j\rangle$ already diagonalizes $\hat{H}_{\rm so}$ (blocks labeled by j, m_j), so the calculation reduces to *effectively non-degenerate* first-order shifts labeled by $j = \ell \pm \frac{1}{2}$.

Time dependent perturbation theory

Many physical phenomena involve interactions that change with time. In atomic and molecular physics, oscillating electromagnetic fields drive transitions between energy levels (absorption, emission, and stimulated emission). In condensed-matter systems, time-dependent perturbations describe the coupling of electrons to phonons, electric fields, or lattice vibrations, and explain phenomena such as optical absorption or spin resonance. In nuclear and particle physics, they account for scattering, decay, and particle excitation under external probes. Time-dependent perturbation theory thus provides a unified framework to compute transition probabilities and rates whenever a system interacts dynamically with its environment, culminating in Fermi's golden rule for transition rates.

Let us now take a general abstract case, where we consider a system described by a Hamiltonian consisting of an unperturbed part and a small *time-dependent perturbation*

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t),$$

with $\hat{H}_0 |n\rangle = E_n |n\rangle$ defining the stationary eigenstates of the unperturbed problem. In the Schrödinger picture, the evolution is governed by

$$i\hbar \frac{d}{dt} |\psi_S(t)\rangle = \hat{H}(t) |\psi_S(t)\rangle .$$

We remove the trivial evolution under $\hat{H}_0$ by defining the interaction-picture state

$$|\psi_I(t)\rangle = e^{\frac{i}{\hbar}\hat{H}_0 t} |\psi_S(t)\rangle .$$

Differentiating and using the Schrödinger equation gives

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle ,$$

where the interaction-picture operator is

$$\hat{V}_I(t) = e^{\frac{i}{\hbar}\hat{H}_0 t} \hat{V}(t) e^{-\frac{i}{\hbar}\hat{H}_0 t}.$$

This is time dynamics with a time dependent Hamiltonian. We have two options now to proceed. One is to recall the Dyson expansion showing time ordering, this we covered shortly in Lecture 2. The second, which is more fun, is to iterate the following integral equation (a formal solution obtained by integrating both sides of the Schrödinger equation)

$$|\psi_I(t)\rangle = |\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1\, \hat{V}_I(t_1) |\psi_I(t_1)\rangle .$$

a few times. For example, writing

$$|\psi_I(t_1)\rangle = |\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^{t_1} dt_2\, \hat{V}_I(t_2)\ |\psi_I(t_2)\rangle .$$

and plugging it back into the formal solution we get

$$\begin{aligned} |\psi_I(t)\rangle &= |\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1\, \hat{V}_I(t_1) \left[|\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^{t_1} dt_2\, \hat{V}_I(t_2)\ |\psi_I(t_2)\rangle \right] \\ &= |\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1\, \hat{V}_I(t_1) |\psi_I(0)\rangle + \left(-\frac{i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2\, \hat{V}_I(t_1)\hat{V}_I(t_2)\ |\psi_I(t_2)\rangle . \end{aligned}$$

Finally, by induction, the exact solution is

$$|\psi_I(t)\rangle = \sum_{n=0}^{\infty}\left(-\frac{i}{\hbar}\right)^n \int_0^t dt_1 \int_0^{t_1} dt_2 \cdots \int_0^{t_{n-1}} dt_n\, \hat{V}_I(t_1)\hat{V}_I(t_2)\cdots\hat{V}_I(t_n)\,|\psi_I(0)\rangle\,.$$

At the end, as the intervals get shorter and shorter $0 \le t_n \le \cdots \le t_2 \le t_1 \le t$, we replaced the $|\psi(t_n\rangle$ with $|\psi(0)\rangle$ Equivalently, introducing the interaction-picture evolution operator $\hat{U}_I(t)$,

$$|\psi_I(t)\rangle = \hat{U}_I(t)\,|\psi_I(0)\rangle\,, \qquad \hat{U}_I(t) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^t dt'\,\hat{V}_I(t')\right],$$

where $\mathcal{T}$ denotes time ordering.

Ok, even if we derived a complicated procedure for estimating the state vector in any order, we will stop the expansion at the first order, which is already sufficient for our derivation of Fermi's golden rule. We move back to the Schrödinger picture

$$|\psi_S(t)\rangle \approx e^{-\frac{i}{\hbar}\hat{H}_0 t}\,|\psi(0)\rangle - \frac{i}{\hbar}\int_0^t dt_1\, e^{-\frac{i}{\hbar}\hat{H}_0 t}\hat{V}_I(t_1)\,|\psi(0)\rangle$$

and sandwich both sides with $\langle m|$

Inserting identities

$$c_m(t) \approx e^{-\frac{i}{\hbar}E_m t}\,c_m(0) - \frac{i}{\hbar}\sum_n \int_0^t dt_1\, e^{-\frac{i}{\hbar}E_m(t-t_1)}\,V_{mn}(t_1)\,e^{-\frac{i}{\hbar}E_n t_1}\,c_n(0)$$

We introduce angular frequencies $\omega_m = E_m/\hbar$. The time evolution $e^{-i\omega_m t}$ is a common factor which we will incorporate into $\tilde{c}_m(t) = c_m(t)e^{i\omega_m t}$ and write

$$\tilde{c}_m(t) \approx c_m(0) - \frac{i}{\hbar}\sum_n \int_0^t dt_1\, e^{i(\omega_m-\omega_n)t_1}\,V_{mn}(t_1)\,c_n(0).$$

Driven time-dependent perturbation. We now consider that the system starts in the initial $|i\rangle$ state, so all coefficients are zero except for $c_i(0) = 1$. We consider an explicitly time-dependent perturbation

$$\hat{V}(t) = \hat{V}\,e^{-i\Omega t} + \hat{V}^\dagger e^{+i\Omega t},$$

which can for example describe the interaction with a classical oscillating field of frequency Ω. Starting from the equation above, we write to first order in the perturbation

$$c_m^{(1)}(t) = -\frac{i}{\hbar}\int_0^t dt'\, e^{i(\omega_m-\omega_i)t'}\,V_{mi}(t').$$

Using $V_{mi}(t') = V_{mi}\,e^{-i\Omega t'} + V_{im}^*\,e^{+i\Omega t'}$, we have

$$c_m^{(1)}(t) = -\frac{i}{\hbar}\int_0^t dt'\,\left[V_{mi}\,e^{i(\omega_m-\omega_i-\Omega)t'} + V_{im}^*\,e^{i(\omega_m-\omega_i+\Omega)t'}\right].$$

The time integration leads to

$$c_m^{(1)}(t) = \frac{V_{mi}}{\hbar}\,\frac{e^{i(\omega_m-\omega_i-\Omega)t}-1}{\omega_m-\omega_i-\Omega} + \frac{V_{im}^*}{\hbar}\,\frac{e^{i(\omega_m-\omega_i+\Omega)t}-1}{\omega_m-\omega_i+\Omega}.$$

To first order, the probability of a transition from $|i\rangle$ to $|m\rangle$ is

$$P_{i\to m}(t) = |c_m^{(1)}(t)|^2 \approx \frac{4|V_{mi}|^2}{\hbar^2}\,\frac{\sin^2[(\omega_m-\omega_i-\Omega)t/2]}{(\omega_m-\omega_i-\Omega)^2} + \frac{4|V_{im}|^2}{\hbar^2}\,\frac{\sin^2[(\omega_m-\omega_i+\Omega)t/2]}{(\omega_m-\omega_i+\Omega)^2}.$$

Each term is peaked when the argument of the sine is stationary, corresponding to

$$\omega_m - \omega_i = \pm\Omega \quad\Longleftrightarrow\quad E_m = E_i \pm \hbar\Omega,$$

that is, absorption $(+)$ or emission $(-)$ of a quantum of energy $\hbar\Omega$.

Fermi's golden rule. In the long-time limit,

$$\frac{\sin^2[(\omega_m - \omega_i \mp \Omega)t/2]}{[(\omega_m - \omega_i \mp \Omega)/2]^2} \xrightarrow[t\to\infty]{} 2\pi t\,\delta(\omega_m - \omega_i \mp \Omega),$$

and the transition rate becomes

$$\Gamma^{(\pm)}_{i\to m} = \frac{2\pi}{\hbar}\,|V_{mi}|^2\,\delta(E_m - E_i \mp \hbar\Omega).$$

The rule says that the rate of the transition between two states depends on the matrix element squared and needs to fulfill a given resonance condition.

Notice that the delta function should be treated as a distribution which peaks the resonance condition when integrated over a continuum of final states.

When many final states $\{|m\rangle\}$ are available, the total rate is the sum over all channels:

$$\Gamma^{(\pm)}_i = \sum_m \Gamma^{(\pm)}_{i\to m} = \frac{2\pi}{\hbar}\sum_m |V_{mi}|^2\,\delta(E_m - E_i \mp \hbar\Omega).$$

Many final states (continuum). For a continuous set of final states,

$$\Gamma^{(\pm)}_i = \frac{2\pi}{\hbar}\int dE\,\rho(E)\,|\langle E|\hat{V}|i\rangle|^2\,\delta\big(E - E_i \mp \hbar\Omega\big) = \frac{2\pi}{\hbar}\,\rho\big(E_i \pm \hbar\Omega\big)\,|\langle E_i \pm \hbar\Omega|\hat{V}|i\rangle|^2.$$

Density of states. The *density of states* $\rho(E)$ is defined such that $\rho(E)\,dE$ gives the number of available quantum states with energies between E and $E + dE$. Formally, for discrete energy levels E_m one can write

$$\rho(E) = \sum_m \delta(E - E_m),$$

so that in the continuum limit the discrete sum becomes

$$\sum_m f(E_m) \;\longrightarrow\; \int dE\,\rho(E)\,f(E).$$

In three-dimensional free-particle systems, for example,

$$\rho(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E},$$

showing that the number of available states grows with $\sqrt{E}$.

Applications of Fermi's golden rule Fermi's Golden Rule is widely used to calculate transition rates between quantum states under a weak perturbation. Its applications span multiple fields of physics:

(i) **Atomic and molecular physics**

- Spontaneous and stimulated emission — rate of photon emission by an excited atom into the electromagnetic field.
- Photoabsorption — transition probability for an atom absorbing a photon.
- Ionization and photodetachment — ejection of an electron by light.
- Collisional excitation or relaxation — transitions between internal levels due to collisions.

(ii) **Condensed matter physics**

- Electron–phonon scattering — determines resistivity and thermal conductivity.

- Electron–impurity scattering — affects carrier mobility in metals and semiconductors.
- Quantum tunneling — current through a potential barrier (e.g. tunnel junction, STM tip).
- Optical absorption — interband electronic transitions induced by light.

(iii) **Nuclear and particle physics**

- Radioactive decay rates — β-decay, α-decay probabilities per unit time.
- Scattering cross sections — connection between transition probabilities and measurable cross sections.
- Particle production — calculation of reaction rates in high-energy collisions.

(iv) **Quantum optics and open systems**

- Spontaneous emission — exponential decay of excited atomic populations.
- Emission into structured reservoirs — decay into cavity or waveguide modes.
- Light–matter coupling — weak-drive transition rates between atomic or molecular levels.

0.1 Appendix: Tensor products for state vectors and operators

As an example, consider a product state $|\psi\rangle = |\phi_{\text{orb}}\rangle \otimes |\chi_{\text{spin}}\rangle$, where

$$|\phi_{\text{orb}}\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad |\chi_{\text{spin}}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Their tensor product is

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ 0 \\ b \\ 0 \end{pmatrix}.$$

The tensor product of operators is computed by multiplying each element of the first matrix by the entire second matrix. For example,

$$\hat{O} = \hat{\sigma}_z \otimes \mathbb{1}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then,

$$\hat{O} = \begin{pmatrix} 1 \cdot \mathbb{1}_2 & 0 \cdot \mathbb{1}_2 \\ 0 \cdot \mathbb{1}_2 & (-1) \cdot \mathbb{1}_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$